



Request for leave: Rajat Roy on 24th March

1 message

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Tue, 24 Mar, 2026 at 18:59

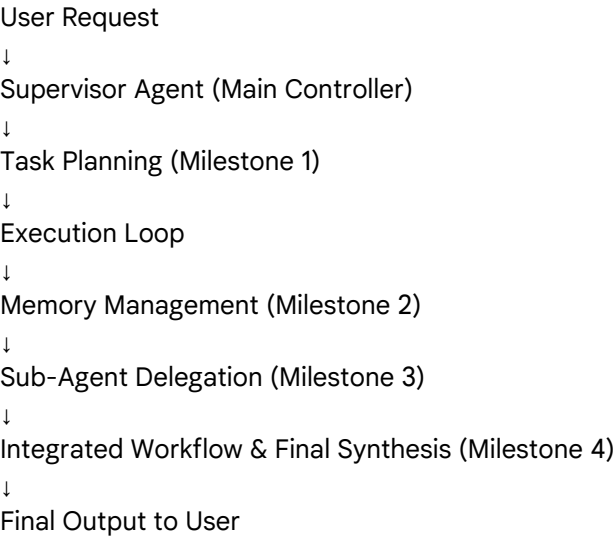
As requested, I am sharing a complete high-level architecture brief updated up to Milestone 4, so that you can clearly understand how all components from earlier milestones connect into one unified workflow.

1. Overall System Objective

The main goal of our project is to build an Autonomous Cognitive Agent that can handle long-horizon tasks such as research generation, comparison workflows, and structured report writing.

Instead of responding in a single step, the system performs structured reasoning through multiple stages. These stages include planning, execution, delegation, memory management, and final synthesis.

At a high level, the full architecture now looks like this:



Each milestone added one critical capability to this pipeline.

2. Milestone 1 – Task Planning Layer

Milestone 1 introduced the planning capability.

When the user provides a complex request, the system does not immediately generate a final answer. Instead, the supervisor agent breaks the request into smaller structured steps using the planning tool.

Example:

User Request:
"Generate a report comparing AI ethics frameworks."

Generated TODO Plan:

1. Identify major AI ethics frameworks
2. Gather relevant details
3. Summarize each framework
4. Compare frameworks

5. Generate final structured report

These TODO tasks are stored in the LangGraph state under the todos section.

This milestone established structured execution thinking.

Without planning, the rest of the architecture cannot function reliably.

3. Milestone 2 – Context Offloading via Virtual File System

Milestone 2 introduced memory management.

During multi-step tasks, intermediate outputs can become large. If everything stays inside the context window, the model may lose earlier information.

To solve this, we introduced a virtual file system inside the agent state.

The file system includes:

- write_file → store intermediate results
- read_file → retrieve stored content
- edit_file → update stored data
- ls → list stored files

Example Workflow:

Step 1 → Gather research data

Step 2 → Summarize research

Step 3 → Store summary in file

Step 4 → Later retrieve file when needed

This allows the agent to maintain structured memory without exceeding context limits.

Milestone 2 transformed the system from a planner into a memory-aware executor.

4. Milestone 3 – Sub-Agent Delegation

Milestone 3 introduced multi-agent collaboration.

Instead of a single agent performing every task, the system now supports specialized sub-agents.

Examples of sub-agents:

- Summarization Agent
- Web Research Agent
- Analysis Agent

The supervisor agent decides whether to perform a task directly or delegate it.

Example:

If a TODO step requires summarization:

Supervisor → Calls task("summarizer")

Summarizer → Processes input

Result → Returned to supervisor

Supervisor → Continues workflow

This creates a team-based architecture where the supervisor coordinates specialized workers.

Milestone 3 transformed the system into a collaborative multi-agent workflow.

5. Milestone 4 – Full Integration & Use Case Execution

Milestone 4 integrates everything into a single cohesive workflow.

At this stage, the system must:

- Generate a structured plan
- Execute tasks sequentially
- Delegate specialized work
- Store intermediate results
- Retrieve stored data
- Combine outputs
- Generate final output

All previous milestones operate together inside one execution loop.

Example End-to-End Flow:

User Request:

"Generate a research report on global AI ethics frameworks."

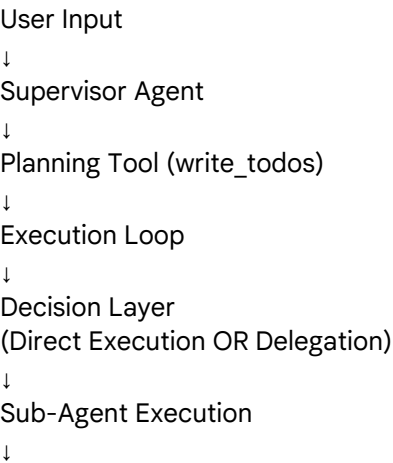
Execution Flow:

1. Planning Phase
Supervisor creates structured TODO plan.
2. Execution Phase
Supervisor executes each TODO step.
3. Delegation Phase
Specialized sub-agents handle specific tasks.
4. Memory Phase
Intermediate outputs stored using write_file.
5. Retrieval Phase
Stored summaries retrieved using read_file.
6. Synthesis Phase
All results combined into final structured output.

This milestone transforms the system into a complete autonomous workflow engine.

6. The Final Integrated Architecture (Milestone 4 View)

Now the system behaves as a fully integrated pipeline:



Memory Storage (write_file)

↓

Memory Retrieval (read_file)

↓

Final Aggregation

↓

Final Output

At this stage, every component depends on every other component.

The system is no longer modular in isolation – it functions as a connected workflow.

7. Evaluation Criteria for Milestone 4

At the end of this milestone, evaluation focuses on end-to-end system performance.

You will test the system using 10–20 complex tasks such as:

- Generate a research report on topic X
- Compare multiple frameworks
- Produce structured analysis documents

Evaluation checks:

1. Task Completion Success

Did the system complete the full workflow without errors?

2. Delegation Behavior

Did specialized tasks use sub-agents correctly?

3. Memory Usage

Were intermediate outputs stored and reused properly?

4. Output Quality

Is the final output logical, structured, and useful?

The success requirement is:

>70% successful task completion with high-quality outputs.

8. Final Architectural Understanding

At Milestone 4 completion, your system is expected to function as:

A planning system

+

A memory system

+

A delegation system

+

An execution system

+

A synthesis system

All working together in one continuous workflow.

This is what makes the system capable of solving long-horizon tasks.

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